Further results on the brief *Unphysicality of Hilbert spaces for quantum mechanics*

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Abstract

This note collects results that go beyond the technical brief *Unphysicality of Hilbert spaces for quantum mechanics* [1], which is based on [2], and partial answers to its open problems. They are not part of the brief. Each section is self-contained, and each names the Lean declarations that check it; the Lean files are in the brief's directory, under `HilbertUnphysicality/FurtherResults/`.

1 [B]COBS fails in one direction only

Condition [B]COBS asks that measurable quantities depend continuously on the state [3]. The brief's Proposition 2 shows that QSTO:H violates it: for the oscillator, the vectors

$$x^j = \sqrt{\tfrac{j-1}{j}}\,|0\rangle + \sqrt{\tfrac{1}{j}}\,|j\rangle$$

converge to $|0\rangle$ in norm, while the expectation of the number operator is $E_N(x^j) = 1$ for every j and $E_N(|0\rangle) = 0$. The jump goes up. This section shows that it always goes up: for an observable that is bounded below, the expectation can exceed its value at the limit, but never fall below it.

The mathematics is classical: it is the statement that the quadratic form of a semibounded self-adjoint operator is closed, which is equivalent to lower semicontinuity (Schmüdgen [5, Prop. 10.1], Simon [6, Thm. 10.14]). What is added here is its reading as a positive, one-sided form of [B]COBS that QSTO:H does satisfy.

1.1 Statement

Let A be a self-adjoint operator on $\mathcal{H}$ with $A \geq c$ for some $c \in \mathbb{R}$, and let E be its spectral measure. For $\psi \neq 0$ put

$$\langle A\rangle_\psi = \frac{1}{\|\psi\|^2}\int_{[c,\infty)} \lambda\, d\langle\psi, E(\lambda)\psi\rangle \in [c, +\infty].$$

It is finite exactly when ψ lies in the form domain $D(|A|^{1/2})$, and it equals $\langle\psi, A\psi\rangle/\|\psi\|^2$ on $D(A)$. Off the form domain it is $+\infty$: such a state has infinite expected energy, which is the physically correct value, rather than none.

Theorem 1.1. *The map $\psi \mapsto \langle A\rangle_\psi$ is lower semicontinuous on $\mathcal{H}\setminus\{0\}$ in the norm topology. That is, if $\psi_k \to \psi \neq 0$, then*

$$\langle A\rangle_\psi \leq \liminf_{k\to\infty} \langle A\rangle_{\psi_k}.$$

Proof. Since $\langle A\rangle_\psi = c + \langle A - c\rangle_\psi$ and adding a constant preserves lower semicontinuity, we may assume $c = 0$. For $n \in \mathbb{N}$ let $A_n = A\,E([0,n])$, a bounded operator. The function $f_n(\psi) = \langle\psi, A_n\psi\rangle/\|\psi\|^2$ is continuous on $\mathcal{H}\setminus\{0\}$, and $f_n(\psi) = \|\psi\|^{-2}\int_{[0,n]} \lambda\, d\langle\psi, E(\lambda)\psi\rangle$ increases to $\langle A\rangle_\psi$ by monotone convergence. A pointwise supremum of continuous functions is lower semicontinuous. □

When A is diagonal in an orthonormal basis (b_i), with eigenvalues $w_i \geq 0$, the same argument reads: $\langle A\rangle_\psi = \|\psi\|^{-2}\sum_i w_i|\langle b_i, \psi\rangle|^2$ is a sum of nonnegative continuous functions of ψ, hence lower semicontinuous. This covers the number operator and the oscillator Hamiltonian, and it is the case checked in Lean.

Corollary 1.2. *For the number operator N, $E_N = \langle N\rangle$ is lower semicontinuous on $\ell^2 \setminus \{0\}$ and agrees with the brief's expectation on $D(N)\setminus\{0\}$. It is not upper semicontinuous at $|0\rangle$: along the brief's sequence,* $\liminf_j E_N(x^j) = 1 > 0 = E_N(|0\rangle)$.

Remark 1.3 (Mixed states). For a density operator ρ put $\operatorname{Tr}\rho A = \sup_n \operatorname{Tr}\rho A_n \in [0,\infty]$ (for $A \geq 0$). Since $|\operatorname{Tr}\rho A_n - \operatorname{Tr}\sigma A_n| \leq n\,\|\rho-\sigma\|_1$, each $\rho \mapsto \operatorname{Tr}\rho A_n$ is continuous in the trace norm, so $\rho \mapsto \operatorname{Tr}\rho A$ is lower semicontinuous on density operators. (Not formalized.)

1.2 Interpretation

A small change of state can add a large amount of energy, by moving a small probability to very high energies; it cannot remove energy. So for semibounded quantities the failure of [B]COBS in QSTO:H is one-sided, and it has a usable consequence: if preparations ψ_k approximate a target state ψ in norm, their energies bound the energy of the target from above in the limit.

Semiboundedness is needed. Position is not semibounded, and its expectation is neither lower nor upper semicontinuous.

Example 1.4. Let φ be the normalized real Gaussian, $\varphi_a(x) = \varphi(x-a)$, and $\psi_k = \sqrt{1-1/k}\,\varphi + \sqrt{1/k}\,\varphi_{a_k}$ with $a_k = \pm k^2$. Then $\psi_k \to \varphi$ in $L^2(\mathbb{R})$ and $\|\psi_k\| \to 1$. The cross terms $\langle\varphi,\varphi_a\rangle$ and $\langle\varphi, X\varphi_a\rangle$ are $O(|a|\,e^{-a^2/4})$, so $\langle X\rangle_{\psi_k} = \pm k + o(1) \to \pm\infty$, while $\langle X\rangle_\varphi = 0$. (Not formalized.)

The theorem also says what a replacement for QSTO:H has to add to obtain full continuity: control of the tails of the state. In the topology of the Schwartz sequences $S(\mathbb{N})$, for example, E_N is continuous (`InconsistentConvergence.SchwartzObservables.continuous_numberExp`).

1.3 Lean

Module `HilbertUnphysicality.FurtherResults.LowerSemicontinuity`, namespace `HilbertUnphysicality.FurtherResults`. Expectations take values in $[0,\infty]$; the general case of Theorem 1.1, which needs the spectral theorem, is not formalized.

- `diagForm`, `lowerSemicontinuous_diagForm`: the form $x \mapsto \sum_i w_i|x_i|^2 \in [0,\infty]$ on $\ell^2(\iota)$, $w_i \geq 0$, is lower semicontinuous (from `lp.lowerSemicontinuous_tsum_mul_enorm_pow`, in `ForMathlib/Analysis/Normed/Lp/Semicontinuous.lean`).
- `diagForm_ne_top_iff`, `diagForm_eq_ofReal`: it is finite exactly on the form domain, and there equals the real sum.
- `diagExpectation`, `lowerSemicontinuousOn_diagExpectation`, `le_liminf_diagExpectation`: the expectation $\langle A\rangle_x$ is lower semicontinuous on $x \neq 0$; Theorem 1.1 for diagonal $A \geq 0$.

- `numberExpectation_eq_ofReal`: for N, it agrees with the brief's `HilbertUnphysicality.expN` on $D(N) \setminus \{0\}$.
- `le_liminf_numberExpectation` and `not_upperSemicontinuousWithinAt_numberExpectation` together give Corollary 1.2.

2 No state has finite position variance in every frame

Proposition 3 of the brief shows that [B]WDQ fails for one change of frame, $y = \tan(\frac{\pi}{2} \operatorname{erf} x)$: the Gaussian packet has finite position variance in the frame x and infinite variance in the frame y. This section asks for [B]WDQ for all smooth changes of frame at once, for the position of one degree of freedom on $L^2(\mathbb{R})$, and shows that no nonzero state survives the free evolution under that demand.

A *smooth change of frame* $y = e(x)$ is a homeomorphism e of $\mathbb{R}$ such that e and e^{-1} are C^∞. Its unitary is $(U_e\psi)(y) = \psi(e^{-1}(y))\,|(e^{-1})'(y)|^{1/2}$, and $U_e\psi$ lies in the domain of the position operator Y exactly when

$$\int e(x)^2 |\psi(x)|^2 \, dx < \infty,$$

that is, when ψ has finite position variance in the frame e.

2.1 All frames force a cutoff

Lemma 2.1. *For every sequence $(c_k)_{k\in\mathbb{N}}$ of reals there is a smooth change of frame e with $|e(x)| \geq c_k$ whenever $|x| \geq k + 1$.*

Proof. Let σ be a smooth monotone transition function, 0 on $(-\infty, 0]$ and 1 on $[1, \infty)$, and put $e(x) = x + \sum_k b_k(\sigma(x - k) - \sigma(-x - k))$ with $b_k \geq 0$. The k-th term vanishes on $[-k, k]$, so the sum is locally finite and e is smooth; each term is monotone, so $e' \geq 1$, e is a bijection of $\mathbb{R}$, and e^{-1} is smooth by the inverse function theorem. For $x \geq k + 1$ the terms $0, \ldots, k$ are complete, so $e(x) \geq x + \sum_{j\leq k} b_j$, and symmetrically for $x \leq -(k + 1)$. Choosing the b_k large enough gives the bound. □

Theorem 2.2. *For $\psi \in L^2(\mathbb{R})$, the following are equivalent:*

1. *ψ has finite position variance in every smooth frame;*
2. *ψ vanishes almost everywhere outside a bounded set.*

Proof. If $\psi = 0$ a.e. outside $[-R, R]$, a continuous e is bounded on $[-R, R]$ and $\int e^2|\psi|^2 < \infty$. Otherwise infinitely many shells $k + 1 \leq |x| < k + 2$ carry mass $m_k = \int_{\text{shell}} |\psi|^2 > 0$. Lemma 2.1 with $c_k = m_k^{-1/2}$ on those shells gives an e with $\int_{\text{shell}} e^2|\psi|^2 \geq 1$ for infinitely many k, so the variance in the frame e is infinite. □

Carcassi, Calderón and Aidala make the same observation for the expectations of all functions of position [2, Appendix, Sec. VIII]. Theorem 2.2 uses only changes of frame, which is what [B]WDQ asks for.

2.2 The free evolution destroys the cutoff

Theorem 2.3. *Let $\psi \in L^2(\mathbb{R})$, $\psi \neq 0$, vanish outside a bounded set, and let $t \neq 0$. Then $e^{-itH_0}\psi$, with $H_0 = -\frac{1}{2}\frac{d^2}{dx^2}$, does not vanish outside any bounded set.*

The statement is well known; the usual proof writes $e^{-itH_0}\psi$ with the free propagator as a chirp times the Fourier transform of a function of bounded support. That needs the Fourier transform of the non-integrable chirp, so the proof below, which is the one checked in Lean, stays on the Fourier side and uses complex analysis instead.

Proof. With the convention $\hat{f}(\xi) = \int e^{-2\pi i x\xi} f(x)\,dx$, the evolution multiplies the Fourier transform by $e^{-ia\xi^2}$, $a = 2\pi^2 t \neq 0$. Suppose ψ and $\phi = e^{-itH_0}\psi$ both vanish outside a bounded set. By Paley–Wiener, $\hat{\psi}$ and $\hat{\phi}$ extend to entire functions G and H of exponential type, and $H(z) = e^{-iaz^2}G(z)$ on $\mathbb{C}$ by the identity theorem; both satisfy $|G(z)|, |H(z)| \leq Ce^{B|z|}$. Choose ω with $\omega^2 = i\operatorname{sign} a$. For real s and u,

$$H(s\omega) = e^{|a|s^2}\,G(s\omega), \qquad H(iu\omega) = e^{-|a|u^2}\,G(iu\omega).$$

So $|G(s\omega)| \leq Ce^{B|s|-|a|s^2}$ and $|H(iu\omega)| \leq Ce^{B|u|-|a|u^2}$. The entire function $k(s) = G(s\omega)H(s\omega)$ is of exponential type, decays faster than any exponential along the positive real axis, and is bounded on the imaginary axis. By the Phragmén–Lindelöf principle in this form, $k = 0$ on the right half-plane. For real $s > 0$, $k(s) = e^{|a|s^2}G(s\omega)^2$, so G vanishes on the ray $\{s\omega : s > 0\}$, hence everywhere by the identity theorem. Then $\hat{\psi} = 0$ and $\psi = 0$. □

Corollary 2.4. *No nonzero $\psi \in L^2(\mathbb{R})$ has finite position variance in every smooth frame both at time 0 and at some time $t \neq 0$ of the free evolution.*

Proof. By Theorem 2.2 both ψ and $e^{-itH_0}\psi$ would vanish outside a bounded set, which Theorem 2.3 excludes. □

The brief's remark that quantum mechanics does not allow a cutoff on all variables gives the companion statement: by Paley–Wiener, no nonzero state has a cutoff on position and on momentum (checked with the brief, `HilbertUnphysicality.quantum_no_cutoff`). So the same demand on position and momentum leaves only $\psi = 0$ even without dynamics.

2.3 Consequence

[B]WDQ, read for all smooth changes of frame, has no nonzero solution that survives the dynamics, in $L^2(\mathbb{R})$ or in any subspace of it. Any replacement for QSTO:H must therefore name the group of changes of frame it supports, and the choice of state space and of frame group are one decision. Antoine [4, Sec. II] already requires the state space of a rigged Hilbert space to be invariant under the representation of a given symmetry group; Corollary 2.4 shows that the group cannot be all changes of frame.

2.4 Lean

Module:
`HilbertUnphysicality.FurtherResults.FrameCutoffs`,
namespace `HilbertUnphysicality.FurtherResults`.

- `smoothFrameChange`, `smoothFrameChange_mem_positionDomain_iff`: the unitary U_e (`Lp.compDiffeo` for $g = e^{-1}$), and $U_e\psi \in D(Y)$ iff $e\psi \in L^2$.

- `Real.exists_contDiff_homeomorph_le_abs`: Lemma 2.1, in `ForMathlib/Analysis/SpecialFunctions/SmoothTransition.lean`.
- `memLp_mul_forall_iff`, `forall_smoothFrameChange_mem_positionDomain_iff`: Theorem 2.2.
- `Real.ae_eq_zero_of_fourier_eq_cexp_mul`, in `ForMathlib/Analysis/Fourier/Chirp.lean`: a chirp on the Fourier side destroys bounded support. It uses `Real.exists_differentiable_fourier_eq` (`ForMathlib/Analysis/Fourier/PaleyWiener.lean`) and Mathlib's `PhragmenLindelof.eq_zero_on_right_half_plane_of_superexponential_decay`.
- `freeEvolution_not_cutoff`: Theorem 2.3, with `MeasureTheory.Lp.fourier_coeFn_eq_of_integrable` (`ForMathlib/Analysis/Fourier/LpSpace.lean`).
- `not_forall_frames_freeEvolution`: Corollary 2.4.

References

[1] G. Carcassi, *Unphysicality of Hilbert spaces for quantum mechanics*, technical brief, Assumptions of Physics, `https://assumptionsofphysics.org/resources/briefs`.

[2] G. Carcassi, F. Calderón, C. A. Aidala, The unphysicality of Hilbert spaces, Quantum Stud.: Math. Found. **12**, 13 (2025), arXiv:2308.06669.

[3] G. Carcassi, T. Thrien, C. A. Aidala, Reverse physics for quantum mechanics, arXiv:2608.27543.

[4] J.-P. Antoine, Dirac formalism and symmetry problems in quantum mechanics. II. Symmetry problems, J. Math. Phys. **10** (1969) 2276–2290.

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[6] B. Simon, Tosio Kato's work on non-relativistic quantum mechanics, arXiv:1711.00528.